

CLASS 10 · ICSE · SELINA CONCISE MATHEMATICS · PYQ SOLUTIONS

Chapter 8: Remainder and Factor Theorems

Step-by-step solutions to the previous year questions

The question numbers match the PYQ question sheet.

1 The factor theorem

1. Which factor is common to $x^2 - 4$ and $x^3 - x^2 - 4x + 4$? (a) $(x + 1)$ (b) $(x - 1)$ (c) $(x - 2)$ (d) $(x - 4)$

ICSE 2025, 1 mark

SOLUTION

$$x^2 - 4 = (x - 2)(x + 2)$$

$$x^3 - x^2 - 4x + 4 = x^2(x - 1) - 4(x - 1) = (x - 1)(x - 2)(x + 2)$$

Of the options, $(x - 2)$ divides both.

Answer: (c)

2 Factorising a cubic polynomial

2. $(x - 2)$ is a factor of $2x^3 - 7x^2 + kx - 2$. Find k , and then factorise the polynomial completely.

ICSE 2025, 3 marks

SOLUTION

By the factor theorem $f(2) = 0$: $16 - 28 + 2k - 2 = 0$, so $k = 7$.

Divide $2x^3 - 7x^2 + 7x - 2$ by $(x - 2)$: the quotient is $2x^2 - 3x + 1$.

$$2x^2 - 3x + 1 = (2x - 1)(x - 1)$$

Answer: $k = 7$; $2x^3 - 7x^2 + 7x - 2 = (x - 2)(2x - 1)(x - 1)$

3. $(x - 1)$ is a factor of $3x^3 + 8x^2 - 15x + k$. Find k , and then factorise the polynomial completely.

ICSE 2024, 3 marks

SOLUTION

$f(1) = 0$: $3 + 8 - 15 + k = 0$, so $k = 4$.

Divide $3x^3 + 8x^2 - 15x + 4$ by $(x - 1)$: the quotient is $3x^2 + 11x - 4$.

$$3x^2 + 11x - 4 = 3x^2 + 12x - x - 4 = (3x - 1)(x + 4)$$

Answer: $k = 4$; $(x - 1)(3x - 1)(x + 4)$

4. Use the factor theorem to factorise $2x^3 - x^2 - 13x - 6$ completely. **ICSE 2023, 4 marks**

SOLUTION

Try $x = -2$: $f(-2) = -16 - 4 + 26 - 6 = 0$, so $(x + 2)$ is a factor.

Dividing by $(x + 2)$ gives the quotient $2x^2 - 5x - 3$.

$$2x^2 - 5x - 3 = 2x^2 - 6x + x - 3 = (2x + 1)(x - 3)$$

Answer: $(x + 2)(2x + 1)(x - 3)$

5. Use the factor theorem to factorise $6x^3 + 17x^2 + 4x - 12$ completely. **ICSE 2020, 3 marks**

SOLUTION

Try $x = -2$: $f(-2) = -48 + 68 - 8 - 12 = 0$, so $(x + 2)$ is a factor.

Dividing by $(x + 2)$ gives the quotient $6x^2 + 5x - 6$.

$$6x^2 + 5x - 6 = 6x^2 + 9x - 4x - 6 = (2x + 3)(3x - 2)$$

Answer: $(x + 2)(2x + 3)(3x - 2)$

6. Use the factor theorem to show that $(x - 2)$ is a factor of $x^3 + x^2 - 4x - 4$. Then factorise the polynomial completely. **ICSE 2019, 3 marks**

SOLUTION

$f(2) = 8 + 4 - 8 - 4 = 0$, so $(x - 2)$ is a factor.

Dividing by $(x - 2)$ gives the quotient $x^2 + 3x + 2 = (x + 1)(x + 2)$.

Answer: $(x - 2)(x + 1)(x + 2)$

7. Use the remainder theorem to factorise $2x^3 + 3x^2 - 9x - 10$. **ICSE 2018, 3 marks**

SOLUTION

Try $x = 2$: $f(2) = 16 + 12 - 18 - 10 = 0$, so $(x - 2)$ is a factor.

Dividing by $(x - 2)$ gives the quotient $2x^2 + 7x + 5 = (2x + 5)(x + 1)$.

Answer: $(x - 2)(x + 1)(2x + 5)$

3 The remainder theorem

8. What must be subtracted from $x^3 + x^2 - 2x + 1$ so that the result is exactly divisible by $(x - 3)$? (a) -31 (b) -30 (c) 30 (d) 31 **ICSE 2024, 1 mark**

SOLUTION

The number to subtract is the remainder on division by $(x - 3)$, which is $f(3)$.

$$f(3) = 27 + 9 - 6 + 1 = 31$$

Answer: (d)

9. What must be added to $2x^3 - 3x^2 - 8x$ so that it leaves a remainder of 10 when divided by $2x + 1$?

ICSE 2020, 3 marks

SOLUTION

Let k be added. The remainder on division by $2x + 1$ is the value at $x = -\frac{1}{2}$.

$$2\left(-\frac{1}{2}\right) - 3\left(\frac{1}{4}\right) - 8\left(-\frac{1}{2}\right) + k = -\frac{1}{2} - \frac{3}{4} + 4 + k = 3 + k$$

$$3 + k = 10$$

Answer: 7

10. Find the remainders when $x^3 + (kx + 8)x + k$ is divided by $x + 1$ and by $x - 2$. The two remainders add up to 1. Find k . ICSE 2019, 3 marks

SOLUTION

$$f(x) = x^3 + kx^2 + 8x + k$$

$$f(-1) = -1 + k - 8 + k = 2k - 9 \text{ and } f(2) = 8 + 4k + 16 + k = 5k + 24$$

$$(2k - 9) + (5k + 24) = 1, \text{ so } 7k = -14$$

Answer: remainders $2k - 9$ and $5k + 24$; $k = -2$

11. What must be subtracted from $16x^3 - 8x^2 + 4x + 7$ so that the result has $2x + 1$ as a factor?

ICSE 2017, 3 marks

SOLUTION

The number to subtract is the remainder on division by $2x + 1$, which is $f\left(-\frac{1}{2}\right)$.

$$f\left(-\frac{1}{2}\right) = 16\left(-\frac{1}{8}\right) - 8\left(\frac{1}{4}\right) + 4\left(-\frac{1}{2}\right) + 7 = -2 - 2 - 2 + 7 = 1$$

Answer: 1

4 Finding unknown constants

12. $(x - 2)$ is a factor of $x^3 - kx - 12$. Find k . (a) 3 (b) 2 (c) -2 (d) -3 ICSE 2023, 1 mark

SOLUTION

$$f(2) = 0: 8 - 2k - 12 = 0, \text{ so } k = -2.$$

Answer: (c)

13. Find a if $(x - a)$ is a factor of $3x^3 + x^2 - ax - 81$. ICSE 2023, 4 marks

SOLUTION

$$\text{By the factor theorem } f(a) = 0: 3a^3 + a^2 - a \times a - 81 = 0$$

$$3a^3 = 81, \text{ so } a^3 = 27.$$

Answer: $a = 3$

14. $(x + 2)$ and $(x + 3)$ are factors of $x^3 + ax + b$. Find a and b . ICSE 2018, 3 marks

SOLUTION

$$f(-2) = 0: -8 - 2a + b = 0 \dots (1)$$

$$f(-3) = 0: -27 - 3a + b = 0 \dots (2)$$

Subtracting (2) from (1): $19 + a = 0$, so $a = -19$. From (1), $b = 8 + 2a = -30$.

Answer: $a = -19$, $b = -30$

These questions are based on questions asked in past board examinations. They have been reworded by MathSetu, the figures are redrawn, and the solutions are MathSetu's own. MathSetu is an independent study resource and is not affiliated with or endorsed by CBSE, CISCE, NCERT or any publisher.

MathSetu